Preprints.org (www.preprints.org) | NOT PEER-REVIEWED | Posted: 29 September 2025 doi:10.20944/preprints202509.2421.v1

no-go: from one covector v_μ no nonzero traceless torsion irrep can be built. This reduces the most general one-derivative ansatz to the $\{\delta^A{}_{[B}\partial_{C]}\epsilon,\ \epsilon^A{}_{BCD}\partial^D\epsilon\}$ span, after which the Palatini equations and the scalar projector select the pure-trace map used in the C1 uniqueness theorem.

Appendix C. Appendix C: Three–Chain Reductions & Improvement Currents (σ_ϵ Scheme)

Scope (σ_ϵ scheme and naming). This appendix provides the paper–checkable reductions behind Section 5 under the *sign-compensated* convention $\mathrm{sgn}(\Sigma_\epsilon)\, I_T \equiv \mathrm{sgn}(\Sigma_\epsilon) I_T = -6\eta^2|\Sigma_\epsilon|$ and the rank-one determinant route naming (formerly "DBI"-type; not Born–Infeld gravity): (i) a rank-one determinant route built out of the canonical traceless matrix $\mathcal{T}^\mu{}_\nu$, (ii) a closed–metric rank–one deformation, and (iii) the PT–even CS/Nieh–Yan shadow. At quadratic order, *each route* reduces in the bulk to the same invariant line

$$\boxed{\delta^2\mathcal{L}_X = A_\star\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T \ +\ \nabla_\mu J^\mu_X, \qquad A_\star = \lambda^2/8,\ \ X \in \{rank-onedeterminantroute, \mathrm{CM}, \mathrm{CS}^+\}\,,}$$

with improvements $\nabla_\mu J^\mu_X$ differing by boundary choices (A4/A5). Closed representatives for J^μ are given on FRW/weak–field backgrounds.

Notation and key relation. We use the preamble shorthands $\Sigma_\epsilon \equiv \Pi_{\mathrm{PT}}[(\partial\epsilon)^2]$, $\mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon)$, $\hat n_\mu \equiv \dfrac{\partial_\mu\epsilon}{\sqrt{|\Sigma_\epsilon|}}$, $\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat n^\mu\hat n_\nu - \frac14\delta^\mu{}_\nu\Big)$, $\tau \equiv \hat n^\mu T_\mu \ = \ 3\,\eta\ \mathrm{sgn}(\Sigma_\epsilon)\sqrt{|\Sigma_\epsilon|}$ so that $\mathrm{Tr}\mathcal{T} = 0$, $\mathrm{Tr}(\mathcal{T}^2) = \frac34$, and (after C1)

$$\tau^2 = 9\eta^2|\Sigma_\epsilon| = \frac{3}{2}\,(-\,\mathrm{sgn}(\Sigma_\epsilon) I_T), \qquad \mathrm{sgn}(\Sigma_\epsilon)\, I_T \equiv \mathrm{sgn}(\Sigma_\epsilon) I_T = -6\eta^2|\Sigma_\epsilon|.$$

Appendix C.1. rank-one determinant route: Determinant Algebra with the $\frac23$ Normalization

Consider the rank-one determinant route Lagrangian

$$\mathcal{L}_{\mathrm{ROD}} = \sqrt{-g}\Big(\sqrt{\det\big[\mathbf{1} + \tfrac23\lambda\,\tau\,\mathcal{T}\big]} - 1\Big), \qquad \lambda \in \mathbb{R}. \tag{A29}$$

Using $\sqrt{\det(\mathbf{1}+X)} = 1 + \frac12\mathrm{Tr}X + \frac18[(\mathrm{Tr}X)^2 - 2\mathrm{Tr}X^2] + \mathcal{O}(X^3)$ and $\mathrm{Tr}\mathcal{T} = 0$, the quadratic piece is

$$\begin{aligned}\delta^2\mathcal{L}_{\mathrm{ROD}} &= \sqrt{-g}\,\frac18\Big[-2\,\mathrm{Tr}\Big((\tfrac23\lambda\,\tau)^2\mathcal{T}^2\Big)\Big] = -\frac14(\tfrac23)^2\lambda^2\sqrt{-g}\,\tau^2\,\mathrm{Tr}(\mathcal{T}^2)\\ &= -\frac14\cdot\frac49\cdot\frac34\,\lambda^2\sqrt{-g}\,\tau^2 = -\frac{1}{12}\,\lambda^2\sqrt{-g}\,\tau^2.\end{aligned} \tag{A30}$$

With $\tau^2 = 9\eta^2|\Sigma_\epsilon|$ and $\mathrm{sgn}(\Sigma_\epsilon)\, I_T = -6\eta^2|\Sigma_\epsilon|$,

$$-\frac{1}{12}\lambda^2\tau^2 = -\frac{1}{12}\lambda^2\,(9\eta^2|\Sigma_\epsilon|) = -\frac34\lambda^2\eta^2|\Sigma_\epsilon| = \Big(\frac{\lambda^2}{8}\Big)\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T,$$

so that

$$\boxed{\delta^2\mathcal{L}_{\mathrm{ROD}} = A_\star\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T \ +\ \nabla_\mu J^\mu_{\mathrm{ROD}}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{A31}$$

At the bulk-density level one may take $J^\mu_{\mathrm{ROD}} \equiv 0$. For unified boundary diagnostics we adopt a common canonical representative J^μ_{can} for *all* three routes (Section C.4).

© 2025 by the author(s). Distributed under a Creative Commons CC BY license.